REMARKS ON MONTGOMERY'S CONJECTURES ON DIRICHLET SUMS

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by

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1. Introduction

This note deals mainly with two conjectures from [M] (p. 72, 73) on the behaviour of general Dirichlet sums

$$\sum_1^N a_n n^{is} \quad ; \quad s = \sigma + it. \tag{1.1}$$

The first conjecture (9.1 in [M]) states in particular that if $\mathcal{T}$ is a 1-separated set of reals in the interval $[-T, T]$, then

$$\sum_{t \in \mathcal{T}} | \sum_1^N a_n n^{it} |^2 \leq C_\epsilon (N + | T |^\epsilon | \mathcal{T} |) \sum_1^N | a_n |^2 \tag{1.2}$$

for all $\epsilon > 0$. Here $\{a_n\}$ are arbitrary scalars ; $| \mathcal{T} |$ is the cardinality of the set $\mathcal{T}$. The distributional inequality derived from (1.2) yields immediately the moment inequality

$$\int_0^T | \sum_1^N a_n n^{it} |^{2\nu} \, dt \leq (NT)^\epsilon (T + N^\nu) (\sum | a_n |^2)^\nu \tag{1.3}$$

for all $\nu \geq 1$. For $1 \leq \nu \leq 2$, (1.3) is a slightly weakened version of conjecture 9.2 of [M]. Taking $\nu = 1$, it is known that

$$\int_{T_0}^{T_0 + T} | \sum_1^N a_n n^{it} |^2 \, dt \leq C(T + N) (\sum_1^N | a_n |^2) \tag{1.4}$$

(see [M], p. 50) and hence (1.3) holds when ν is an integer. It is also shown in [M] that a statement such as (1.3) implies the density conjecture for the zero's of the Riemann zeta function $\zeta(s)$ in particular, i.e.

$$N(\sigma, T) << T^{2(1-\sigma)+\epsilon} \tag{1.5}$$

(see [M], p. 100).

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As will be observed here, both conjectures 1.2 and 1.3 are to strong and at least some restrictions on the coefficients $\{a_n \mid 1 \leq n \leq N\}$ have to be made. H. Montgomery proposed recently following conjecture (*) $(\log T \sim \log N)$

$$\sum_{t \in \mathcal{T}} \mid \sum_{1}^{N} a_n n^{it} \mid^2 << N^{\epsilon}(N + \mid \mathcal{T} \mid)N \tag{1.6}$$

assuming $\mid a_n \mid \leq 1$. As far as the zero-density estimates concerns, (1.6) applies equally well. Conjecture (1.6) will also be discussed here. More precisely, we will derive from (1.6) certain function theoretic inequalities, implying in particular statements about the Hausdorff dimension of so-called Kakea sets in higher dimension : If (1.6) holds, then following is true :

$$\text{any set in } \mathbb{R}^d,\ d \geq 2, \text{ containing a translate ofevery line, has dimension } d \tag{1.7}$$

This fact is true for $d = 2$ but open for $d \geq 3$. This last question belongs to an interesting problem area in geometric measure theory which is also of relevance to Fourier analysis.

Assuming the Lindelöf hypothesis, one has the estimate

$$\mid \sum_{1}^{N} n^{it} \mid << \frac{N}{\mid t \mid} + T^{\epsilon} N^{1/2} \tag{1.8}$$

implying (1.2) if $a_n = 1$, $1 \leq n \leq N$, $a_n = 0$ for $n > N$. On the other hand (see [M], p. 66), if we let

$$M(\alpha, T) = \max_{\substack{\sigma \geq \alpha, \mid t \mid \leq T \\ \mid s - 1 \mid \geq 1}} \mid \zeta(s) \mid \tag{1.9}$$

then one may write following bound on the left member of (1.2)

$$\sum_{t \in \mathcal{T}} \mid \sum_{N}^{2N} a_n n^{it} \mid^2 << (N + N^{\alpha} M(\alpha, 4T) \mid \mathcal{T} \mid)(\sum_{N}^{2N} \mid a_n \mid^2). \tag{1.10}$$

Thus, under the Lindelöf hypothesis, $\mid \mathcal{T} \mid$ would carry a factor $(NT)^{\epsilon}.N^{1/2}$ in (1.10). This inequality turns out to be essentially best possible for general ℓ^2-coefficients. Observe that (1.6) amounts to bound the Dirichlet operator(*) $(n^{it})_{\substack{N \leq n \leq 2N \\ t \in \mathcal{T}}}$ from $\ell^{\infty}_{[N,2N]}$ to $\ell^2_{\mathcal{T}}$ rather than from $\ell^2_{[N,2N]}$ to $\ell^2_{\mathcal{T}}$. Unfortunately, the methods used in [M] or [B] are Hilbertian (estimates of $\| D \|$ by $\| D^* D \|^{1/2}$) and it seems that new ideas are needed to make the distinction between ℓ^{∞} and ℓ^2-domain of D.

(*) In the sence of [B], p. 32.

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Finally, it is observed that in (1.6) the factor N^{ϵ} has to appear in front and not only as coefficient of $\mid \mathcal{T} \mid$. In fact, we show that for $0 < \rho < 1$ there is a Dirichlet sum $S(s)$ of the form (1.1) with $\mid a_n \mid \leq 1$ and a separated set $\mathcal{T} \subset [0, N^2]$ for which

$$\sum_{t \in \mathcal{T}} \mid S(it) \mid^2 > \exp(\log N)^{\rho} (N + \mid \mathcal{T} \mid .N^{\epsilon(\rho)})^N \tag{1.11}$$

where $\epsilon(\rho) > 0$. Improving this example seems a combinatorial problem.

2. Short sums

Fix N, $N < T < N^2$ and let k be a positive integer such that

$$T.\frac{k^2}{N^2} = o(1) \tag{2.1}$$

(in fact, less than a suitable numerical constant).

For $0 \leq r \leq k$, $n = N + r$ and $\mid t \mid \leq T$, write

$$n^{it} = N^{it} \left(1 + \frac{r}{N}\right)^{it} = N^{it} e^{i\frac{r}{N}t} e^{it[\log(1+\frac{r}{N}) - \frac{r}{N}]}$$

where the last factor is of the form

$$1 + \sum_{\substack{j \geq 1 \\ j' \geq 2j}} c_{jj'} t^j \left(\frac{r}{N}\right)^{j'} = 1 + \sum_{\substack{j \geq 1 \\ j' \geq 0}} c'_{jj'} \left(\frac{t}{T}\right)^j \left(T\frac{k^2}{N^2}\right)^j \left(\frac{r}{k}\right)^{2j} \left(\frac{r}{N}\right)^{j'}.$$

Standard considerations permit then to estimate

$$\sum_{r=0}^{k} a_{N+r} e^{i\frac{r}{N}t} \tag{2.2}$$

by expressions of the form

$$\sum_{I} a_n n^{it}$$

where I runs over subintervals of $[N, N+k]$ (k was choosen such that on $[N, N+k]$ one may linearize $\log n$)

Take

$$\left.\begin{array}{l} a_n = 1 \text{ if } N \leq n \leq N + k \\ = 0 \text{ otherwise} \end{array}\right\} \tag{2.3}$$

and

$$\mathcal{T} = \bigcup_{q=0,1,\cdots,[\frac{T}{2\pi N}]} (\mathcal{T}_1 + 2\pi N q) \tag{2.4}$$

where $\mathcal{T}_1$ is a 1-separated set in the interval $\left[0, \frac{N}{100k}\right]$. Thus

$$| \mathcal{T} | \sim \frac{T}{k} \tag{2.5}$$

and

$$F(t) \equiv \sum_{r=0}^{k} e^{i\frac{r}{N}t} \tag{2.6}$$

is $\approx k$ for $t \in \mathcal{T}$.

The left member of (1.2) is of order $k^2 \mid \mathcal{T} \mid \sim kT$ and hence

$$> c\left\{(\frac{T}{N})N + k \mid \mathcal{T} \mid\right\} \sum \mid a_n \mid^2 . \tag{2.7}$$

For a choice $T = N^{1+\delta}$, $0 < \delta < 1$, (2.7) equals

$$c(N^{1+\delta} + N^{\frac{1-\delta}{2}} \mid \mathcal{T} \mid) \sum \mid a_n \mid^2 . \tag{2.8}$$

In particular, (1.2) fails and the optimal bound one may hope for is essentially $N\left(1 + T^{\epsilon}N^{1/2} \mid \mathcal{T} \mid\right) \sum \mid a_n \mid^2$, implied by the Lindelöf hypothesis.

Choosing $1 < \nu < 2$ and putting now

$$T = N^{\nu} \tag{2.9}$$

one also have

$$\int_0^T \mid F(t) \mid^{2\nu} dt \geq \int_{\Omega} \mid F(t) \mid^{2\nu} dt \geq T \cdot k^{2\nu-1} = T \cdot k^{\nu-1}(\sum \mid a_n \mid^2)^{\nu} \tag{2.10}$$

where

$$\Omega = \bigcup_{0 \leq q \leq [\frac{T}{2\pi N}]} \left[0, \frac{N}{100k}\right] . \tag{2.11}$$

Hence, also (1.3) fails for $1 < \nu < 2$.

REMARK 2.12. For $\infty > p > 2$, call a 1-separated set $\Lambda \subset \mathbb{R}$ a Λ_p-set of constant $K \geq 1$, provided the inequality

$$\left(\int_0^1 \mid \sum_{\lambda \in \Lambda} a_\lambda e^{i\lambda t} \mid^p\right)^{1/p} \leq K \left(\sum_{\lambda \in \Lambda} \mid a_\lambda \mid^2\right)^{1/2} \tag{2.13}$$

holds, for all scalar sequence $(a_\lambda)_{\lambda\in\Lambda}$. Letting $T = N^\nu$ in (1.3) and making a change of variable to normalize the interval of integration, (1.3) means that the set

$$\{T \cdot \log n \mid 1 \leq n < N\} \tag{2.14}$$

is a $\Lambda_{2\nu}$ set of constant N^ϵ. It is an elementary fact that such sets do not contain arithmetic progressions (or $\frac{1}{2}$-perturbations of arithmetic progressions) longer than $\sim N^{\epsilon/\nu-1}$. Now the set $\{T \cdot \log(N+r) \mid 0 \leq r \leq k\}$ where k is given by (2.1) is a perturbation of the progression $T \cdot \log N + \frac{T \cdot r}{N}$.

On the other hand, the inequality (cf. [M])

$$\int_0^{N^2} |\sum_1^N a_n n^{it}|^4 \, dt \leq C N^2 \sum_1^N |a_n(2)|^2 \tag{2.15}$$

derived from (1.4), yields that the set

$$\Lambda = \bigcup_{k=1}^{\infty} \{4^k \log p \mid 2^k < p < 2^{k+1}, \quad p \text{ prime}\} \tag{2.16}$$

is a $\Lambda(4)$-set of bounded constant (and no $\Lambda(q)$-set for any $q > 4$). This is perhaps the simplest construction of such sets.

3. The modified conjecture (1.6)

This conjecture states a bound on the operator $\mathcal{D} = \{n^{it}\}_{\substack{1 \leq n \leq N \\ t \in \mathcal{T}}}$ acting on $\ell^\infty_{[1,n]}$, $n\ell$

$$\left(\sum_{t\in\mathcal{T}} |\sum_1^N a_n n^{it}|^2\right)^{1/2} << N^\epsilon (N + |\mathcal{T}|) N. \max_{1\leq n\leq N} |a_n| \tag{3.1}$$

Our purpose is to derive a function theoretic consequence of an estimate of the form

$$\left(\sum_{t\in\mathcal{T}} |\sum_1^N a_n n^{it}|^2\right)^{1/2} \leq (\alpha(N)N + \beta(N) \cdot |\mathcal{T}|) \cdot N \cdot \max_{1\leq n\leq N} |a_n| \tag{3.2}$$

for $\mathcal{T}$ a 1-separated set in $[0, N^2]$. Here $\alpha(N)$, $\beta(N)$ stand for slow growing functions of N. We will show the following

LEMMA 3.3. *Assume (3.2) holds. Let $R, B \geq 1$ and $0 \leq f \leq 1$ a function supported by the interval $[0, B]$ such that $|f'| < R$. Then*

$$\int_0^1 \sup_\tau \left[\sum_{b=1}^B f(\tau + bx)\right] dx \leq C(\alpha(R^2B) + \beta(R^2B) \cdot \int f). \tag{3.4}$$

This inequality will be used to derive the dimension result for Kakea sets mentioned above and to obtain (1.11). Let $N \leq T \leq N^2$ and k be as above, i.e.

$$Tk^2 \sim N^2. \tag{3.5}$$

By (a_n) we always mean a 1-bounded sequence of scalars. Split $[N, 2N]$ in intervals I of length k. Clearly (3.2) implies following "square function" inequality

$$\sum_{t\in\mathcal{T}} \sum_{I} \mid \sum_{n\in I} a_n n^{it} \mid^2 < (\alpha \cdot N + \beta \cdot \mid \mathcal{T} \mid)N. \tag{3.6}$$

Using the linear approximation to $\log n$ on the intervals I, the considerations in section 2 yield that

$$\sum_{t\in\mathcal{T}} \sum_{M}^{2M} \mid \sum_{r=0}^{k-1} a_{mk+r} e^{i\frac{r}{mk}t} \mid^2 < C(\alpha N + \beta \mid \mathcal{T} \mid)N \tag{3.7}$$

where

$$M = \frac{N}{k}. \tag{3.8}$$

For each $m = M,\ M+1, \cdots, 2M$, fix a point $t_m \in \mathbb{R}$. Denoting

$$D(x) = \sum_{0}^{k-1} e^{ijx} \tag{3.9}$$

the Dirichlet kernel, a specification of the coefficients a_n in (3.7) gives

$$\sum_{t\in\mathcal{T}} \sum_{M}^{2M} \mid D_k\left(\frac{t - t_m}{mk}\right) \mid^2 < C(\alpha N + \beta \mid \mathcal{T} \mid)N. \tag{3.10}$$

Making a change of variable $t = kt'$, it follows that

$$\sum_{t'\in\mathcal{T}'} \sum_{M}^{2M} \mid D_k\left(\frac{t' - t'_m}{m}\right) \mid^2 < C(\alpha N + \beta \mid \mathcal{T} \mid)N \tag{3.11}$$

where $\mathcal{T}'$ is an arbitrary $\frac{1}{k}$-separated set in $\left[0, \frac{T}{k}\right]$.

Consider a set $\mathcal{A}$ of 1-separated points in $\left[0, \frac{T}{k}\right]$ and put

$$\mathcal{T}' = \sum_{t\in\mathcal{A}} \left(t + \{\frac{j}{k} \mid 0 \leq j < k\}\right) \tag{3.12}$$

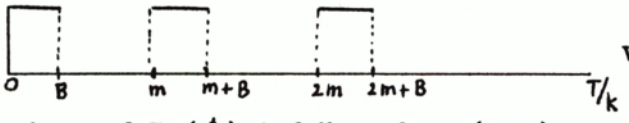
which is a $\frac{1}{k}$-separated set of size

$$| \mathcal{T}' | = k \cdot | \mathcal{A} | . \tag{3.13}$$

Since $k << M$, $D_k(\frac{t}{m})$ is approximatively constant on intervals of length 1. It follows from (3.12), (3.13) that

$$\sum_{t \in \mathcal{A}} \sum_{M}^{2M} | D_k \left(\frac{t - t_m}{m} \right) |^2 < C(\alpha \cdot M + \beta \cdot | \mathcal{A} |)N. \tag{3.14}$$

Consider the function χ_m, 0,1-valued, defined as



where $B = \dfrac{M}{k}$.

Because of the shape of $D_k(\frac{t}{m})$, it follows from (3.14)

$$\sum_{t \in \mathcal{A}} \sum_{M}^{2M} \chi_m(t - t_m) < C(\alpha M + \beta \cdot | \mathcal{A} |)B. \tag{3.15}$$

Consider the case where $\mathcal{A}$ is a union of intervals of length B in $\left[0, \frac{T}{k}\right] \cap \mathbb{Z}$. Thus

$$\mathcal{A} = \mathcal{A}' + [0, B] \tag{3.16}$$

where the points of $\mathcal{A}'$ are B-separated. Since

$$| (\mathcal{A}' + [0, B] - t_m) \cap (m\mathbb{Z} + [-B, B]) | \geq \frac{B}{2} \; | \; (\mathcal{A}' + [0, B] - t_m) \cap m\mathbb{Z} \; |$$

it follows from (3.15) that for such $\mathcal{A}$

$$\sum_{t \in \mathcal{A}} \sum_{M}^{2M} y_m(t - t_m) < C(\alpha M + \beta \; | \; \mathcal{A} \; |) \tag{3.17}$$

where y_m denotes the indicator function of $m.\mathbb{Z}$.

Rewrite (3.17) as

$$\sum_{M}^{2M} \sum_{j \in \mathbb{Z}} \chi_{\mathcal{A}}(t_m + jm) < C(\alpha \cdot M + \beta \cdot | \mathcal{A} |). \tag{3.18}$$

Let now f be the indicator function of a set in $[0, B]$ which is a union of intervals of length $\frac{1}{k} = \frac{B}{M}$.

On has

$$\begin{aligned} \int_1^2 \sup_\tau \left(\sum_1^B f(\tau + bx) \right) dx &= \sum_M^{2M} \int_{[\frac{n}{M}, \frac{m+1}{M}]} \sup_\tau \left(\sum_1^B f(\tau + bx) \right) dx \\ &\approx \frac{1}{M} \sum_M^{2M} \sum_1^B f(\tau_m + b\frac{m}{M}) \end{aligned} \tag{3.19}$$

because of the assumption on f and for some choice of $\tau_m = \frac{t_m}{M}$. The set

$$\mathcal{A} = \left\{ t \in \mathbb{Z},\ \ 0 \leq t \leq M.B \mid f(\frac{t}{M}) = 1 \right\} \tag{3.20}$$

is a union of B-intervals and is contained in $[0, \frac{T}{k}]$, since $T \sim M \cdot B \cdot k$. Thus, by (3.18), (3.19) is bounded by

$$C\left(\alpha + \beta \cdot \frac{|\mathcal{A}|}{M} \right) \sim C\left(\alpha(k^2 B) + \beta(k^2 B) \cdot \int_0^B f \right). \tag{3.21}$$

Statement (3.4) is now clear, k corresponding to the derivative estimate R.

4. Consequences to the Hausdorff dimension of Kakea sets

We assume (1.6), i.e. (3.4) holds taking $\alpha(N) = \beta(N) = C_\epsilon N^\epsilon$, $\epsilon > 0$ a fixed arbitrarily small number. Then the following statement holds

> Let A be a subset of $\mathbb{R}^d$, $d \geq 2$ such that every line has a translate contained in A. Then $\dim A = d$. (4.1)

Here dim stands for Hausdorff dimension. As mentioned earlier, this is a fact for $d = 2$ (see [F] for details).

The main point consists in replacing (3.4) by its higher dimensional analogue. The argument is routine. This will permit us to derive a lower estimate for metrical entropy numbers.

Assume A a union of subcubes of $[0, B]^d$ of size $\frac{1}{N}$. Assume that for every $x \in [0,1]^d$. There is some $\tau(x) \in \mathbb{R}^d$ such that

$$\tau(x),\ \tau(x) + x, \cdots, \tau(x) + Bx \in A.$$

Consider the standard partition of $\mathbb{R}_+^d$ in cubes of size $\frac{1}{N}$ which are labeled by a multi-index $(j_1, \cdots, j_d) \in \mathbb{Z}_+^d$. Put $R = N^d$. Define $Y_{(j_1,\cdots,j_d)}$ as the interval $(j_1 + j_2 N + \cdots + j_d N^{d-1})\frac{1}{R} + [0, \frac{1}{R}]$. For $j \in \mathbb{Z}$, let $\tilde{j}$ denotes its residue mod N. Let J stand for the $(j_1, \cdots, j_d)$ corresponding to a cube contained in A. Thus

$$| J | = R \cdot \mathrm{mes}(A). \tag{4.2}$$

Consider following subset of $[0,1]$

$$S = \bigcup_{(j_1,\cdots,j_d)\in J} Y_{(\tilde{j}_1,\cdots,\tilde{j}_d)}. \tag{4.3}$$

Thus S is a union of $\frac{1}{R}$-intervals of measure

$$\mathrm{mes}(S) \leq \mathrm{mes}(A). \tag{4.4}$$

Define the set $F \subset [0, B]$

$$F = S + \left\{ \frac{q_0}{N^d} + \frac{q_1}{N^{d-1}} + \frac{q_2}{N^{d-2}} + \cdots + q_d \mid q_i \in \mathbb{Z}, \ \ | q_i | < B \right\} \tag{4.5}$$

for which by (4.4)

$$\mathrm{mes}(F) < (C.B)^d \mathrm{mes}(A). \tag{4.6}$$

Assume $y \in Y_{(j_1,\cdots,j_d)} \subset [0,1]$. By hypothesis

$$\tau\left(\frac{j_1}{N}, \cdots, \frac{j_d}{N}\right) + b\left(\frac{j_1}{N}, \cdots, \frac{j_d}{N}\right) \in A \quad \text{for} \quad 0 \leq b \leq B.$$

Thus for some $j_1', \cdots, j_d' \in \mathbb{Z}_+$,

$$(j_1' + bj_1, \cdots, j_d' + bj_d) \in J \ , \quad 0 \leq b \leq B.$$

Consequently, for $0 \leq b \leq B$

$$S \supset \frac{(j_1' + bj_1)^\sim}{N^d} + \frac{(j_2' + bj_2)^\sim}{N^{d-1}} + \cdots + \frac{(j_d' + bj_d)^\sim}{N} + [0, \frac{1}{R}] \tag{4.7}$$

and therefor, by construction of F

$$F \supset \left(\frac{j_1'}{N^d} + \frac{j_2'}{N^{d-1}} + \cdots + \frac{j_d'}{N}\right) + b\left(\frac{j_1}{N^d} + \frac{j_2}{N^{d-1}} + \cdots + \frac{j_d}{N}\right) + \left[\frac{-B}{R}, \frac{B}{R}\right]. \tag{4.8}$$

Since $y \in Y_{(j_1,\cdots,j_d)}$,

$$\left(\frac{j_1'}{N^d} + \frac{j_2'}{N^{d-1}} + \cdots + \frac{j_d'}{N}\right) + by \in F \ \text{ for } \ 0 \leq b \leq B. \tag{4.9}$$

Apply (3.4) with $f = \chi_F$. If $B > C_\epsilon R^{3\epsilon}$, it follows that

$$\int_0^B f > \frac{1}{C_\epsilon} R^{-3\epsilon}. \tag{4.10}$$

Hence, by (4.6)

$$\text{mes}(A) > (C_\epsilon R^{3\epsilon})^{-d} = R^{-\epsilon'}. \tag{4.11}$$

From the preceeding, a detailed deduction of (4.1) is now straightforward.

REMARK. (4.1) is an open problem for $d = 3$. Presently, I do know A needs to have dimension $\geq \frac{7}{3}$.

5. Growth of the functions $\alpha(N)$, $\beta(N)$ in (3.2).

Lemma 3.3 also permits us to get lower bounds on the numbers $\alpha(N)$, $\beta(N)$. This is pursued next. The method is of combinatorial nature.

Let $S \subset [0,1] \times [0,1]$ be a union of $(\frac{1}{R} \times \frac{1}{R})$-squares. Denote for $\varphi, \psi \in \mathbb{Z}_+$

$$P_{(\varphi,\psi)}(S) = \{\varphi x + \psi y \mid (x,y) \in S\}. \tag{5.1}$$

Let $1 \leq q \leq B$ and $\mathcal{B} \subset \{0,1,\cdots,B\}$. If S satisfies

$$P_{(1,0)}(S) = [0,1] \tag{5.2}$$

then one has

$$|\mathcal{B}| \leq C \left\{ \alpha(R^2 B) + \beta(R^2 B) \sum_{b \in \mathcal{B}} \text{mes}(P_{(b,q)}(S)) \right\}. \tag{5.3}$$

Indeed, let $f = \chi_A$ where

$$A = \bigcup_{b \in \mathcal{B}} P_{(b,q)}(S). \tag{5.4}$$

By (5.2) one may associate to each $x \in [0,1]$ a point τ' s. t. $(x,\tau') \in S$, hence $bx + q\tau' = bx + \tau \in A$ for all $b \in \mathcal{B}$. The left member of (3.4) is thus at least $|\mathcal{B}|$ which yields inequality (5.3).

We now make following construction. Let a and K be positive integers to be specified. Define

$$\xi = (1, -\frac{1}{a}) \qquad \eta = (\frac{1}{a}, 1) \tag{5.5}$$

$$S = \left\{ \sum_{j=0}^{K-1} \frac{x_j \xi + y_j \eta}{a^{2j+1}} \mid x_j, y_j \in \{0, 1, \cdots, a-1\} \right\} \cup [0, a^{-2K}]^2. \tag{5.6}$$

Thus

$$R = a^{2K} \tag{5.7}$$

and

$$P_{(\varphi,\psi)}(S) = \left\{ \sum_{j=0}^{K-1} \frac{(\varphi x_j + \psi y_j)a + (\varphi y_j - \psi x_j)}{a^{2(j+1)}} \right\} + [0, (\varphi + \psi)a^{-2K}].$$

In particular

$$P_{(1,0)}(S) = \left\{ \sum_{j=0}^{K-1} \frac{a x_j + y_j}{a^{2(j+1)}} \right\} + [0, a^{-2K}] = [0, 1] \tag{5.8}$$

and for φ, ψ given by

$$\varphi = ca + d, \quad \psi = -c + da \quad ; \quad c, d \text{ positive integers} < a^{\sigma} \quad (\sigma < 1)$$

$$P_{(\varphi,\psi)}(S) \subset \left\{ (1 + a^2) \sum_{j=0}^{K-1} \frac{c x_j + d y_j}{a^{2(j+1)}} \right\} + [0, a^{-2(K-1)}] \tag{5.9}$$

implying

$$\text{mes}\, P_{(\varphi,\psi)}(S) \leq (2a^{1+\sigma})^K a^{-2(K-1)} < R^{-\frac{1-\sigma}{3}} \tag{5.10}$$

if K is sufficiently large.
Observe also that the equation

$$\frac{ca + d}{-c + da} = \frac{c'a + d'}{-c' + d'a} \tag{5.11}$$

implies

$$(cd' - c'd)a^2 = c'd - cd'$$

hence

$$\frac{c}{d} = \frac{c'}{d'}. \tag{5.12}$$

We will let $d = 1$ and $\mid c \mid < a^{\sigma}$ vary.

LEMMA 5.13. *Given a number U and $0 < \gamma < 1$, there is a set Ω of positive integers of size $\sim U$ and with common multiple q such that*

$$\mid \Omega \mid > U^{\gamma} \tag{5.14}$$

and

$$q < exp(log\, U)^{\frac{1}{1-\gamma}}. \tag{5.15}$$

Proof: Choose prime $p = p_1$ such that

$$\log U \sim p^{1-\gamma} \tag{5.16}$$

and r with

$$10^r p^r \leq U < 10^{r+1} p^{r+1}. \tag{5.17}$$

Let $q_0 = p_1 p_2 \cdots p_s$ be the product of $s \sim \frac{p}{\log p}$ consecutive primes and Ω_0 the set of all simple products x of r primes among $p_1, \cdots, p_s$. Each such x is less than U, by (5.17), and may thus be multiplied with a divisor of $\prod_{p' < p} p'$ to obtain $x' \sim U$. The set Ω of those numbers admit $q = \prod_{p' < p_s} p'$ as common multiple and

$$| \Omega | = | \Omega_0 | = \binom{s}{r}. \tag{5.18}$$

It follows from (5.16), (5.17) that $r \log p \sim p^{1-\gamma}$, hence $\frac{s}{r} \sim p^{\gamma}$ and (5.14). Also (5.15) holds by (5.16). ∎

Assume $\Omega \subset [U, 10U]$. Split the interval $[U, 10U]$ in $\sim U^{1-\sigma}$ intervals of the form $[a - a^{\sigma}, a]$ $(\sigma < 1)$. For $0 < \gamma < 1$ and

$$\sigma > 1 - \gamma \tag{5.19}$$

one may find such an interval $[a - a^{\sigma}, a]$ containing a subset Ω' of Ω of size at least

$$| \Omega' | > U^{\gamma - 1 + \sigma}. \tag{5.20}$$

Defining S as in (5.6), and putting

$$\mathcal{B} = \left\{ (ca+1) \frac{q}{a-c} \mid a - c \in \Omega' \right\} \tag{5.21}$$

it follows from (5.3), (5.7), (5.20)

$$U^{\gamma - 1 + \sigma} \leq \alpha(a^{4K+2} q) + \beta(a^{4K+2} q) \sum_{a - c \in \Omega'} \frac{q}{a-c} \operatorname{mes} P_{(ca+1, a-c)}(S). \tag{5.22}$$

By (5.10), (5.15), one deduces from (5.22)

$$\begin{aligned} U^{\gamma - 1 + \sigma} \leq\ & \alpha((20U)^{4K} \exp(\log U)^{\frac{1}{1-\gamma}}) + \\ & \beta((20U)^{4K} \exp(\log U)^{\frac{1}{1-\gamma}}) . \exp(\log U)^{\frac{1}{1-\gamma}} \cdot U^{-\frac{1-\sigma}{2} K}. \end{aligned} \tag{5.23}$$

Choosing

$$K \sim \frac{1}{1-\sigma} (\log U)^{\frac{\gamma}{1-\gamma}} \tag{5.24}$$

one gets

$$U^{\gamma-1+\sigma} \leq \alpha\left(\exp\frac{1}{1-\sigma}(\log U)^{\frac{1}{1-\gamma}}\right) + \beta\left(\exp\frac{1}{1-\sigma}(\log U)^{\frac{1}{1-\gamma}}\right)\cdot\exp\left(-(\log U)^{\frac{1}{1-\gamma}}\right) \tag{5.25}$$

or letting

$$N \sim \exp\frac{1}{1-\sigma}(\log U)^{\frac{1}{1-\gamma}} \tag{5.26}$$

$$\exp[(\gamma-1+\sigma)(1-\sigma)(\log N)^{1-\gamma}] \leq \alpha(N) + \beta(N)N^{-(1-\sigma)}. \tag{5.27}$$

Here γ, σ are subject to (5.19). Thus one sees that for each $0 < \rho < 1$ there is $\epsilon(\rho) > 0$ such that either $\alpha(N) > \exp(\log N)^{\rho}$ or $\beta(N) > N^{\epsilon(\rho)}$. This corresponds to (1.11).

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